

How To Assess ELECTROMAGNETIC RADIATION From MOBILEPHONE TOWERS



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Over the last few years, there have been enormous installations of telecommunication towers around the world. A mobilephone has evolved at a very fast pace from an uncommon, expensive, brick-shaped object used in some western countries to a very convenient and ubiquitous part of human life, with more than five billion subscriptions worldwide. The rapid expansion of this mass telecom technology has been accompanied by some public and media concern about the possible adverse health issues resulting from exposure to electromagnetic (EM) fields, such as those radio frequencies emitted by antennae installed on telecommunication towers.

As the mobilephone base and various wireless technologies such as Global System for Mobile (GSM) communication, Code Division Multiple Access (CDMA), High-Speed Downlink Packet Access (HSDPA), Worldwide Interoperability for Microwave Access (WiMAX), Wireless Broadband (WiBro), Evolution Data Optimized (EV-DO Advanced) and Long Term Evolution (LTE-Advanced) are rapidly expanding and evolving, requirement for mobilephone towers is also growing proportionately. It is therefore high time that a strict regulatory body is put in place to monitor, control and provide suitable directives for mobile tower installations. Otherwise, such mushrooming of telecom towers could create a sense of panic among general public.

There is a perception regarding the existence of a high level of EM radiation in the vicinity of these towers, which may cause adverse biological effects. Numerous studies have considered the evidence and recommended more research on this burning issue. According to World Health Organization (WHO), INTERPHONE (a 13-country coordinated case-control study started in 2000), Independent Expert Group on Mobile Phones (IEGMP) and Scientific Committee on Emerging and Newly Identified Health Risks (SCENIHR) study researches, it has been found that EM radiation can contribute to health deficiency, including increased risk

of brain tumours, eye cancers, salivary glands tumours, testicular cancers and leukaemia.

Several surveys have found a variety of self-reported symptoms for people who live close to base stations. Collectively, they have not provided evidence of a relationship, but have had sufficient limitations to leave the question unresolved. International Commission on Non-Ionizing Radiation Protection (ICNIRP) study has concluded that exposure levels due to mobilephone base stations are generally around one-ten-thousandth of the guideline levels. Moreover, WHO has classified mobilephone radiation on International Agency for Research on Cancer (IARC) scale into Group 2B (possibly carcinogenic to humans). This means that there could be some risk of carcinogenicity, so additional research into the long-term, heavy use of mobilephones/wireless technologies needs to be conducted.

There is a need for not only controlling the haphazard installation of towers, but also undertaking systematic study for measurement of the radiation levels in some selected high population density urban areas to ensure that power density levels do not exceed the prescribed threshold limits in the frequency range of interest.

A typical telecom system

A wireless telecom system consists of two major parts: base station and receiver. A base station consists of a telecom tower along with indoor electronics and provides coverage to a very large area. A receiver may be a mobile phone, USB device, separate antenna (indoor or outdoor equipment) or PCMCIA card built in the laptop or computer.

A basic telecommunications base station infrastructure model is shown in Fig. 1. It has several entities such as sectorised antenna system, control and communication electronics, transmitters (Trx cards), media converters (electrical to optical and vice-versa), transmission equipment, power supply module and so on.

The main function of a base station is to

Electromagnetic radiation from mobilephone towers is a subject of utmost importance and requires a deep understanding of assessment methods of electromagnetic radiation from these telecommunication installations